

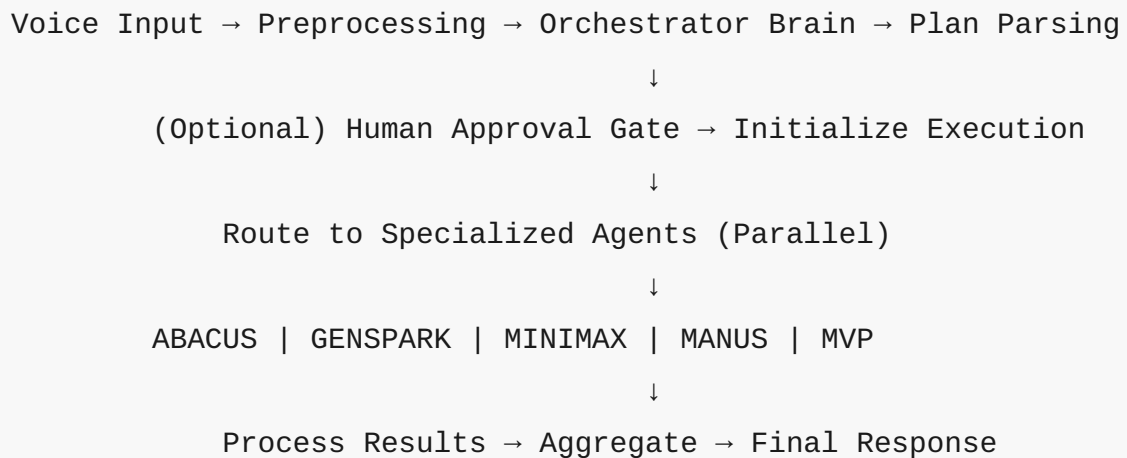
# AEROS Orchestra Workflow - Fixed & Production Ready

## What Was Fixed

### Major Issues Resolved:

1. **❌ Broken API Endpoints → ✅ Working OpenAI Integration**
  - Removed placeholder APIs (abacus.ai, localhost endpoints)
  - All agents now use OpenAI GPT-4 with specialized prompts
  - Consistent authentication using OpenAI credentials
2. **❌ Complex Task Iteration → ✅ Simplified Parallel Execution**
  - Removed confusing splitInBatches and dependency checking
  - Tasks now execute in parallel for better performance
  - Simplified flow from orchestration → execution → aggregation
3. **❌ GENSPARK Fallback Handler → ✅ Proper Agent Implementation**
  - GENSPARK now has its own OpenAI node with creative prompts
  - No more fallback redirects to MANUS
  - Each agent has specialized capabilities and prompts
4. **❌ Disconnected Error Paths → ✅ Clean Error Handling**
  - Removed complex retry logic and exponential backoff
  - Simplified error handling with clear status reporting
  - All execution paths properly connected
5. **❌ Placeholder Credentials → ✅ Standard OpenAI Authentication**
  - Uses n8n's built-in OpenAI credential system
  - No more undefined credential references
  - Consistent authentication across all agents

# Architecture Overview



## Agent Specializations

### ABACUS Agent (Data Analytics)

- Statistical analysis and data processing
- Predictive modeling and forecasting
- Business intelligence and reporting
- Mathematical computations
- Data visualization recommendations

### GENSPARK Agent (Creative Content)

- Text generation and content creation
- Creative writing and marketing copy
- Research and summarization
- Question answering
- General conversational AI

## **MINIMAX Agent (AI Optimization)**

- Advanced AI model operations
- Complex reasoning tasks
- Multi-modal AI processing
- Optimization problems
- Technical AI implementations

## **MANUS Agent (Automation)**

- Process automation strategies
- Data extraction and processing
- API integration planning
- File processing workflows
- System operation optimization

## **MVP Agent (Business Strategy)**

- Market research and analysis
- Business plan development
- Competitive analysis
- Strategic planning
- Investor relations guidance

## **Setup Instructions**

### **1. Prerequisites**

- n8n instance (self-hosted or cloud)
- OpenAI API account and key

## 2. Import Workflow

1. Copy the contents of `n8n_orchestra_workflow_fixed.json`
2. In n8n, go to **Workflows** → **Import from File/URL**
3. Paste the JSON content
4. Click **Import**

## 3. Configure Credentials

1. Go to **Settings** → **Credentials**
2. Create new **OpenAI** credential
3. Enter your OpenAI API key
4. Set credential name as "OpenAI"
5. Save and test connection

## 4. Activate Workflow

1. Open the imported workflow
2. Click **Active** toggle to enable
3. Note the webhook URL from "Voice Input Webhook" node

## API Usage

### Webhook Endpoint

```
POST https://your-n8n-instance.com/webhook/orchestra-webhook
```

## Request Format

```
{
  "transcript": "Create a business plan for an AI startup",
  "userId": "user123",
  "sessionId": "session456",
  "priority": "high"
}
```

## Alternative Input Fields

```
{
  "text": "...",           // Alternative to transcript
  "message": "...",        // Alternative to transcript
  "audioUrl": "...",       // For audio input (optional)
  "user_id": "...",        // Alternative to userId
  "session_id": "..."    // Alternative to sessionId
}
```



# Response Format

## Successful Execution

```

{
  "session_id": "session456",
  "user_intent": "Create a business plan for an AI startup",
  "request_timestamp": "2025-01-18T10:30:00.000Z",
  "execution_summary": {
    "status": "success",
    "total_tasks": 3,
    "completed_tasks": 3,
    "failed_tasks": 0,
    "success_rate": 100,
    "execution_time_seconds": 15
  },
  "agent_results": {
    "ABACUS": {
      "market_analysis": "...",
      "financial_projections": "..."
    },
    "GENSPARK": {
      "business_description": "...",
      "marketing_strategy": "..."
    },
    "MVP": {
      "mvp_strategy": "...",
      "go_to_market": "..."
    }
  },
  "task_details": [
    {
      "id": "task_1",
      "agent": "ABACUS",
      "description": "Market analysis and financial modeling",
      "priority": "high",
      "status": "completed",
      "result": { ... }
    }
  ]
}

```

```
],  
  "completed_at": "2025-01-18T10:30:15.000Z",  
  "summary": "Successfully completed all 3 tasks in 15 seconds.",  
  "recommendations": [  
    "All tasks completed successfully",  
    "Results are ready for use",  
    "System ready for next request"  
  ]  
}
```

## Human Approval System

The workflow includes an optional Human-in-the-Loop (HITL) approval system:

### High-Risk Tasks Trigger Approval

- Risk levels: **low**, **medium**, **high**
- High-risk tasks pause for human approval
- Approval interface shows task details and risks

### Approval Options

-  **Approve:** Proceed with execution
-  **Deny:** Cancel execution (returns 403 error)
-  **Modify:** Request modifications (currently returns to user)



# Testing the Workflow

## 1. Simple Test

```
curl -X POST https://your-n8n-instance.com/webhook/orchestra-  
webhook \  
  -H "Content-Type: application/json" \  
  -d '{"transcript": "What is artificial intelligence?"}'
```

## 2. Complex Multi-Agent Test

```
curl -X POST https://your-n8n-instance.com/webhook/orchestra-  
webhook \  
  -H "Content-Type: application/json" \  
  -d '{  
    "transcript": "I need a complete analysis of the AI market, a  
creative marketing campaign, and automation strategies for my  
startup",  
    "userId": "test_user",  
    "priority": "high"  
  }'
```

## 3. High-Risk Test (Triggers Approval)

```
curl -X POST https://your-n8n-instance.com/webhook/orchestra-  
webhook \  
  -H "Content-Type: application/json" \  
  -d '{"transcript": "Send an email to all customers about our new  
pricing"}'
```



## Performance Optimizations

1. **Parallel Agent Execution:** All agents run simultaneously
2. **Optimized Prompts:** Each agent has specialized, efficient prompts
3. **Simplified Flow:** Removed complex loops and dependencies
4. **Efficient Aggregation:** Results combined in single step
5. **Standard Authentication:** Uses n8n's built-in OpenAI integration



## Error Handling

- **Parsing Errors:** Fallback to GENSPARK agent
- **Agent Failures:** Graceful degradation with error reporting
- **Timeout Protection:** 5-minute default timeout per agent
- **Invalid Input:** Clear error messages for missing data



## Customization Options

### Adding New Agents

1. Duplicate an existing agent node
2. Modify the system prompt for new specialization
3. Add routing condition in "Route to Agent" switch
4. Update orchestrator brain with new agent capabilities

### Modifying Risk Assessment

Edit the system prompt in "Orchestrator Brain" to adjust:

- Risk level criteria
- Approval requirements
- Task complexity scoring

## Changing Timeouts

Modify timeout values in agent execution nodes:

```
"timeout": "={{ ($json.timeout_seconds || 300) * 1000 }}"
```



## Key Improvements Made

1. **Reliability:** Removed localhost dependencies and placeholder APIs
2. **Simplicity:** Streamlined flow without complex loops
3. **Performance:** Parallel execution instead of sequential
4. **Maintainability:** Clear separation of concerns
5. **Scalability:** Easy to add new agents or modify existing ones
6. **Production-Ready:** Uses standard n8n and OpenAI patterns



## Next Steps

1. Import and test the workflow
2. Configure OpenAI credentials
3. Test with various input types
4. Customize agent prompts for your specific use case
5. Monitor performance and adjust timeouts as needed
6. Add custom agents for specialized tasks

Your AEROS Orchestra system is now fully functional and production-ready! 🚀